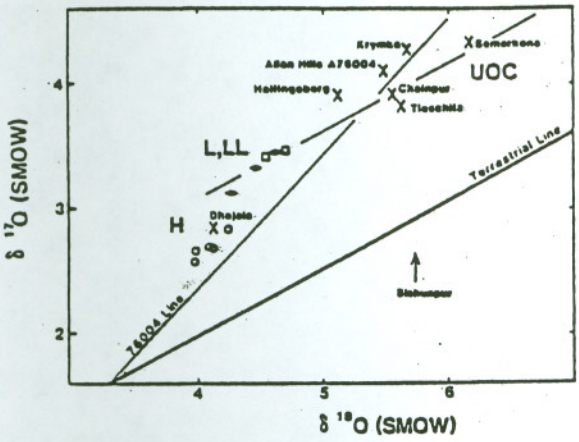
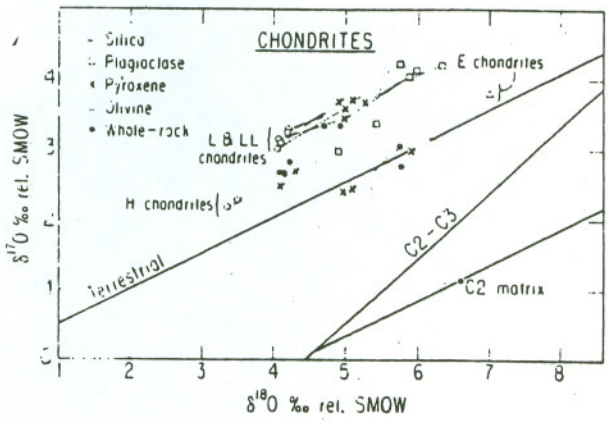
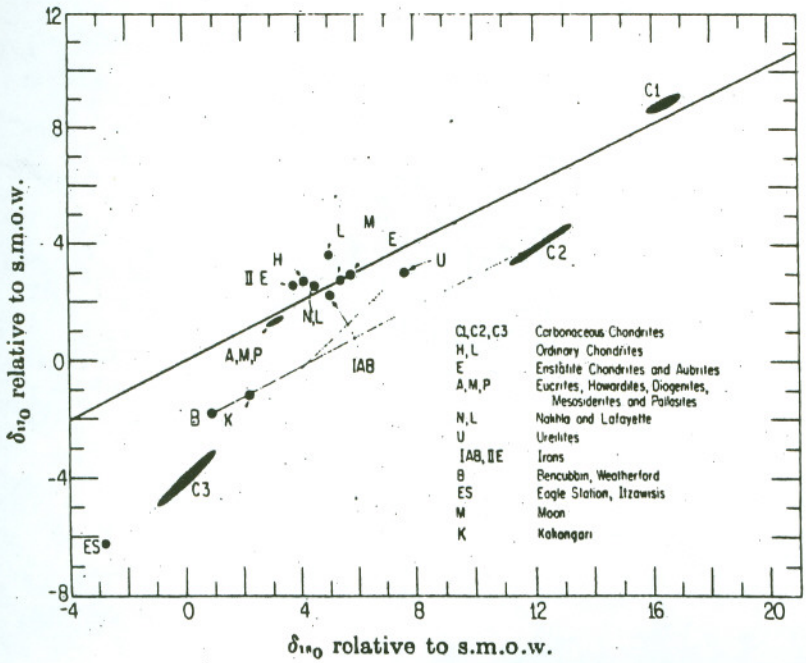
ANALYTICAL CUME
APRIL 4, 2003

The distribution of oxygen isotopes in meteorites is often discussed in a variety of contexts. Write an essay on the topic of oxygen isotopes in cosmochemistry using the data shown in the three diagrams below.





Three-isotope plot for ordinary chondrites and the terrestrial mass fractionation line. Data for equilibrated meteorites (circles, H chondrites; squares, L chondrites; diamonds, LL chondrites) calculated from Clayton et al. [1976] assuming the following percent modal abundances: olivine-H 35, L 48, LL 58; pyroxene-H 25, L 23, LL 16; plagioclase-all classes 10. Data for type 3 chondrites (crosses) from Clayton et al. [1981] and for Allan Hills A76004 from Mayeda et al. [1980].